Solution of Tutorial Sheet-6

(1) real seq^n $\{a_n\}$, where $a_n = r^n$

Case (1) If $r > 1$.

Then for any $h > 0$ & $r = (1+h)$, we have

$$(1+h)^n \geq 1+nh$$

but $1+nh > M$

$$\Rightarrow n > \frac{(M-1)}{h}$$

$\Rightarrow \forall M, \exists n_0 \in \mathbb{N}$ s.t.

$$r^n = (1+h)^n \geq 1+nh > M$$

when $n > n_0$

Hence $\{r^n\}$ is not bounded & hence $\{r^n\}$ doesnot converge.

Case (2) If $r = 1$ then $\{r^n\} \longrightarrow 1$.

Case (3) If $-1 < r < 1$ i.e. $|r| \leq 1$. Let $r \neq 0$

Then for some $h > 0$, we have

$$|r| = \frac{1}{1+h}$$

& $|a_n| = |r^n| = |r|^n = \frac{1}{(1+h)^n} \leq \frac{1}{(nh+1)}$

Now $\frac{1}{(nh+1)} < \varepsilon \Rightarrow nh+1 > \frac{1}{\varepsilon}$

$$\Rightarrow n > \frac{(1/\varepsilon - 1)}{h}$$

Therefore when $n > \frac{(1/\varepsilon - 1)}{h}$, we have

$$|a_n| < \varepsilon.$$

Case ④ If $r = -1$ then oscillates. In otherwords not convergent.

Case ⑤ If $r < -1$

then $t = -r > 1$.

then $a_n = (-1)^n t^n$

Again we will show that $|a_n|$ is not bounded.

$|a_n| = |t|^n > M$ for all $n > N$

Hence real seq^n convergent when

$-1 < r \leq 1$

② Radius of ~~convergence~~ convergence

(i) $\sum_{n=1}^{\infty} \frac{\lfloor n}{n^3} z^n$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\lfloor n+1 / (n+1)^3}{\lfloor n / n^3}\right|$$

$$= \left|\frac{\cancel{(n+1)} \cdot n^3}{(n+1)^{\cancel{3}2}}\right|$$

$$= \left|\frac{n^3}{(n+1)^2}\right| \longrightarrow \infty \text{ as } n \to \infty$$

$\Rightarrow$ Radius of convergence $R = 0$

(ii) $R = \dfrac{1}{\lim_{n\to\infty} |a_n|^{1/n}}$ $\left(\text{for } \sum \left(\dfrac{2n+1}{6n-10}\right)^n (z-1)^n\right.$

$= \dfrac{1}{\lim_{n\to\infty} \left(\dfrac{2n+1}{6n-10}\right)}$ i.e. $\underline{|z-1| < 3}$.

$= 6/2 = 3$ #

(iii) $\sum (-1)^n \dfrac{n}{(n^2+3)} z^n$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-1)^{n+1}\dfrac{(n+1)}{(n+1)^2+3}}{(-1)^n \dfrac{n}{n^2+3}}\right|$$

$$= \left|\frac{(n+1)\,(n^2+3)}{((n+1)^2+3)\,(n)}\right|$$

$$= \left|\frac{\cancel{n^3}(1+1/n)\,(1+3/n^2)}{\cancel{n^3}\left[(1+1/n)^2+3/n^2\right]}\right|$$

as $\underline{n\to\infty}$.

$= 1$

i.e. Radius of convergence is $\underline{|z-0|<1}$

(iv) $\sum k^n z^n$

$R = \dfrac{1}{\lim_{n\to\infty} |a_n|^{1/n}}$

$= \dfrac{1}{\lim_{n\to\infty} (k^n)^{1/n}} = \dfrac{1}{k}$

i.e. $\underline{|z-0| < 1/k}$.

(v) $\sum \frac{5^{n^2}}{n} z^n$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\frac{5^{(n+1)^2}}{(n+1)}}{5^{n^2}/n}\right|$$

$$= \frac{5^{(n+1)^2} \cdot n}{(n+1)\, 5^{n^2}}$$

$$= \frac{\cancel{5^{n^2}} \cdot 5^{(2n+1)} \cdot n}{(n+1)\, \cancel{5^{n^2}}}$$

$$= \frac{5^{(2n+1)}\, n}{(n+1)}$$

$$= \frac{5^{(2n+1)}}{(1+1/n)}$$

$\longrightarrow \infty$ as $n \to \infty$

Hence $\underline{\underline{R = 0}}$

(vi) $\sum_{n=1}^{\infty} \frac{2^n}{n(n+1)} z^{n+2}$

or $\sum_{n=3}^{\infty} \frac{2^{n-2}}{(n-2)(n-1)} z^n$

$$\left|\frac{2^{n+1}/(n+1)(n+2)}{2^n/n(n+1)}\right| = \frac{2\, n(n+1)}{(n+1)(n+2)} \longrightarrow 2 \text{ as } n \to \infty$$

i.e. $\underline{\underline{R = 1/2}}$

(vii) $\sum \frac{z^n}{n^n}$

$$R = \frac{1}{\lim_{n\to\infty} |a_n|^{1/n}}$$

$$= \frac{1}{\lim_{n\to\infty} \frac{1}{n}}$$

$$\underline{\underline{R = \infty}} \quad \text{i.e.} \quad \underline{\underline{|z-0| < \infty}}$$

$$\boxed{\begin{aligned} \cosh z &= \frac{e^z + e^{-z}}{2} \\ \sinh z &= \frac{e^z - e^{-z}}{2} \end{aligned}}$$

(3) (i) $\cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{\lfloor 2n}$ $\quad \underline{\underline{|z| < \infty}}$

$\Rightarrow \cosh(z^2) = \sum_{n=0}^{\infty} \frac{z^{4n}}{\lfloor 2n}$ $\quad \underline{\underline{|z| < \infty}}$

Now $z \cosh(z^2) = \sum_{n=0}^{\infty} \frac{z^{4n+1}}{\lfloor 2n}$ $\quad \underline{\underline{|z| < \infty}}$

(ii) $\frac{z}{z^4+9} = f(z)$

$$= \frac{z}{9} \cdot \frac{1}{1+\left(\frac{z^4}{9}\right)}$$

$$= \frac{z}{9}\left[1 - \left(\frac{z^4}{9}\right) + \left(\frac{z^4}{9}\right)^2 - \cdots\right]$$

$$= \cancel{\frac{z}{9}\left[1 - \left(\frac{z}{\sqrt{3}}\right)^4 + \right] z}$$

$$= \sum (-1)^n \frac{z^{4n+1}}{3^{2n+2}}$$

& radius of convergence is given by

$$\left|\left(\frac{z}{\sqrt{3}}\right)^4\right| < 1$$

$$\Rightarrow \left|\frac{z^4}{9}\right| < 1$$

$$\Rightarrow |z|^4 < 9$$

$$\Rightarrow \boxed{|z| < \sqrt{3}}$$

(4) $f(z) = \frac{1}{1-z}$ about $\underline{\underline{z = i}}$

$$= \frac{1}{(1-i)-(z-i)}$$

$$= \frac{1}{(1-i)\left[1 - \left(\frac{z-i}{(1-i)}\right)\right]}$$

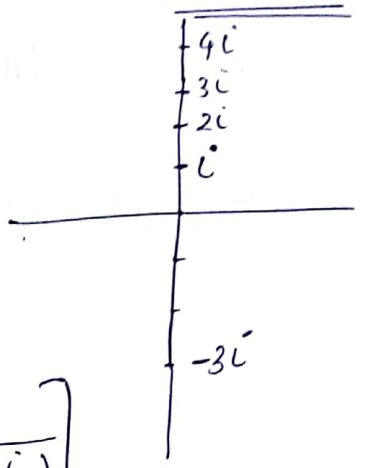
$$= \frac{1}{(1-i)}\left[1 + \frac{(z-i)}{(1-i)} + \frac{(z-i)^2}{(1-i)^2} + \cdots\right]$$

& radius of convergence $\left|\frac{z-i}{1-i}\right| < 1$

$$\Rightarrow |z-i| < |1-i|$$

$$\Rightarrow |z-i| < \sqrt{2}$$

(8) $f(z) = \dfrac{1}{z^2+2iz+3}$ about $z=0$, which converges at $z=4i$



$$= \frac{1}{(z-i)(z+3i)}$$

$$= \frac{1}{4i}\left[\frac{1}{(z-i)} - \frac{1}{(z+3i)}\right]$$

$$= \frac{1}{4i}\left[\frac{1}{z(1-i/z)} - \frac{1}{z(1+\frac{3i}{z})}\right]$$

$|z| > 3$

$$= \frac{1}{4i}\left[\frac{1}{z}\left\{1+\left(\frac{i}{z}\right)+\left(\frac{i}{z}\right)^2 - \cdots\right\} - \frac{1}{z}\left\{1-\frac{3i}{z}+\left(\frac{3i}{z}\right)^2 - \cdots\right\}\right]$$

$|i/z| < 1$, $\left|\frac{3i}{z}\right| < 1$

$1 < |z|$, $3 < |z|$

Therefore it will converges for $3 < |z|$.

(11) Discuss the convergence of the series

$$\sum_{n=-\infty}^{\infty} 3^{-|n|} z^{2n}$$

$\Rightarrow$ We can write the above series like this $\rightarrow$

$$\sum_{n=-\infty}^{\infty} 3^{-|n|} z^{2n} = \sum_{n=0}^{\infty} 3^{-n} z^{2n} + \sum_{n=-\infty}^{-1} 3^{n} z^{2n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{z^2}{3}\right)^n + \sum_{n=1}^{\infty} \left(1/3z^2\right)^n$$

1st series is geometric series & converges when $|z|^2 < 3$ i.e. $\underline{\underline{|z| < \sqrt{3}}}$ & it diverges $\forall$ z with $\underline{\underline{|z| \geq \sqrt{3}}}$.

Similarly, second series converges when

$$\left|\frac{1}{3z^2}\right| < 1$$

i.e. $\frac{1}{3} < |z|^2$

i.e. $\boxed{|z| > \frac{1}{\sqrt{3}}}$

& diverges when $\boxed{|z| \leq \frac{1}{\sqrt{3}}}$

So, it follows that,

$$\sum_{n=-\infty}^{\infty} 3^{-|n|} z^{2n} = \frac{1}{1 - z^2/3} + \frac{1/3z^2}{1 - (1/3z^2)}$$

$$= \frac{8z^2}{(3-z^2)(3z^2-1)}$$

$\forall$ z s.t. $\underline{\underline{\frac{1}{\sqrt{3}} < |z| < \sqrt{3}}}$, whereas one series diverges for all remaining value of z.

(9) Let $a \in (-1, 1)$. Then

$$\frac{a}{(z-a)} = \sum_{n=1}^{\infty} \frac{a^n}{z^n}, \quad \underline{|a| < |z| < \infty}$$

$$\frac{a}{(z-a)} = \frac{a}{z\left[1 - \frac{a}{z}\right]}$$

$$= \frac{a}{z}\left[1 + \left(\frac{a}{z}\right) + \left(\frac{a}{z}\right)^2 + \cdots\right]$$

$$= \sum_{n=1}^{\infty} \left(\frac{a}{z}\right)^n = \sum_{n=1}^{\infty} \frac{a^n}{z^n}$$

$\left|\frac{a}{z}\right| < 1$

$\Rightarrow \boxed{|z| > |a|}$

(10) $f(z) = 1/z$

Then $f(z) = \dfrac{1}{(z-1)+1}$

$$= \frac{1}{1 + (z-1)}$$

$$\frac{1}{z} = 1 - (z-1) + (z-1)^2 - (z-1)^3 + \cdots$$

$$\frac{1}{z} = 1 - (z-1) + (z-1)^2 - (z-1)^3 + \cdots$$

$$\frac{1}{z} = \sum_{n=0}^{\infty} (-1)^n (z-1)^n$$

Differentiate $\rightarrow$

$$-\frac{1}{z^2} = \sum_{n=1}^{\infty} (-1)^n n (z-1)^{n-1}$$

$$\cancel{\frac{1}{z^2} = \sum_{n=1}^{\infty} (-1)^n n (z-1)^{n-1}}$$

$$\frac{1}{z^2} = \sum_{n=0}^{\infty} (-1)^n (n+1) (z-1)^n \qquad \underline{|z-1| < 1}$$

(7) (i) $f(z) = \dfrac{e^z}{(z+1)}$ about $\underline{\underline{z=-1}}$

$$= \frac{e^{(z+1-1)}}{(z+1)} \qquad \underline{\underline{0 < |z+1|}}$$

$$= \frac{e^{-1}\, e^{(z+1)}}{(z+1)}$$

$$= \frac{1}{e}\frac{1}{(z+1)}\left[1 + (z+1) + \frac{(z+1)^2}{\lfloor 2} + \cdots\right]$$

$$= \frac{1}{e}\left[\frac{1}{(z+1)} + 1 + \frac{(z+1)}{\lfloor 2} + \frac{(z+1)^2}{\lfloor 3} + \cdots\right]$$

(ii) $f(z) = \dfrac{\sinh(1/z)}{z^2}$ at $\underline{\underline{z=0}}$, $\left(\sinh z = \dfrac{e^z - e^{-z}}{2}\right)$

$$= \frac{1}{z^2}\left[\frac{1}{z} + \frac{1}{z^3 \lfloor 3} + \frac{1}{z^5 \lfloor 5} + \cdots\right]$$

$$\underline{\underline{0 < |z| < \infty}}$$

$$= \frac{1}{z^3} + \frac{1}{z^5 \lfloor 3} + \frac{1}{z^7 \lfloor 5} + \cdots$$

(iii) $e^{z/z-2}$ at $\underline{\underline{z=2}}$

$$= e^{\frac{z-2+2}{(z-2)}}$$

$$= e^{\left[1 + \frac{2}{(z-2)}\right]}$$

$$= e\left[e^{2/z-2}\right] \qquad \underline{\underline{0 < |z-2| < \infty}}$$

$$= e\left[1 + \frac{2}{(z-2)} + \left(\frac{2}{z-2}\right)^2 \frac{1}{\lfloor 2} + \cdots\right]$$

(iv) $f(z) = \dfrac{1+2z^2}{z^3+z^5}$ $\quad \underline{z=0}$

$= \dfrac{1+2z^2}{z^3[1+z^2]}$ $\quad \underline{0<|z|<1}$

$= \dfrac{(1+2z^2)}{z^3}\left[1 - z^2+z^4-z^6+\cdots\right]$

$= \left(\dfrac{1}{z^3}+\dfrac{2}{z}\right)\left[1 - z^2+z^4-z^6+\cdots\right]$

$= \dfrac{1}{z^3}\left[1-z^2+z^4-z^6+\cdots\right] + \dfrac{2}{z}\left[1-z^2+z^4-z^6+\cdots\right]$

$= \left[\dfrac{1}{z^3} - \dfrac{1}{z} + z - z^3 + \cdots\right] + \dfrac{2}{z} - 2z + 2z^3 - 2z^5 + \cdots$

$= \dfrac{1}{z^3} + \dfrac{1}{z} - z + z^3 - \cdots$ #

(6) find out taylor's series $\longrightarrow$

(i) $f(z) = e^z$, $\quad \underline{z=1}$ $\quad \underline{|(z-1)| < \infty}$

$= e^{z-1+1}$

$= e\, e^{(z-1)}$

$= e\left[1+(z-1)+\dfrac{(z-1)^2}{\lfloor 2} + \cdots\right]$

(ii) $f(z) = \dfrac{1}{(z-2i)}$ $\quad z_0 = 0$

$= \dfrac{1}{(-2i)\left[1-\left(\dfrac{z}{2i}\right)\right]}$ $\quad \left|\dfrac{z}{2i}\right| < 1 \Rightarrow \boxed{|z| < 2}$ #

$= \dfrac{1}{(-2i)}\left[1+\left(\dfrac{z}{2i}\right)+\left(\dfrac{z}{2i}\right)^2+\cdots\right]$

(iii) $\ln(2+iz) \qquad z=i$

$= \ln(1+iz-i^2)$

$= \ln[1+i(z-i)]$

$= \ln[1+\omega] \qquad \underline{|\omega|<1}$, where $\omega = i(z-i)$

$= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \omega^n$

$= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \left(i(z-i)\right)^n$

(iv) $f(z) = \cosh z$ at $\underline{z = i\pi}$

$\cosh z = \frac{e^z + e^{-z}}{2}$

$\cosh z = \left(1 + \frac{z^2}{\lfloor 2} + \frac{z^4}{\lfloor 4} + \cdots\right)$

Note that $\cosh(z+\pi i) = -\cosh z$

i.e. $\cosh z = -\cosh(z+\pi i)$

i.e. period of $\cosh z$ is $\underline{2\pi i}$.

Therefore we have,

$\cosh z = -\cosh(z+\pi i)$

$= -\cosh(z-\pi i)$

Since we know the presentation of $\cosh z \rightarrow$

$\cosh z = 1 + \frac{z^2}{\lfloor 2} + \frac{z^4}{\lfloor 4} + \cdots \qquad |z|<\infty$

$\Rightarrow \cosh(z-\pi i) = -\left[1 + \frac{(z-\pi i)^2}{\lfloor 2} + \frac{(z-\pi i)^4}{\lfloor 4} + \cdots\right]$

$$\cosh(z) = -\left[\sum_{n=0}^{\infty} \frac{(z-\pi i)^{2n}}{\lfloor 2n}\right] \qquad |z-\pi i|<\infty$$

Similarly, we can figure out —

$$\sinh z = -\sum_{n=0}^{\infty} \frac{(z-\pi i)^{2n+1}}{\lfloor (2n+1)} \qquad |z-\pi i|<\infty$$

$$\sinh z = \frac{e^{z}-e^{-z}}{2}$$

$$\sinh z = z + \frac{z^3}{\lfloor 3} + \frac{z^5}{\lfloor 5} + \cdots \qquad |z|<\infty$$

$\sinh(z+\pi i) = -\sinh z$ & period of $\sinh z$ is $2\pi i$. Then we have.

$\sinh z = -\sinh(z+\pi i) = -\sinh(z-\pi i)$

Now we know $\sinh z = z + \frac{z^3}{\lfloor 3} + \frac{z^5}{\lfloor 5} + \cdots$ $\quad |z|<\infty$

$$\Rightarrow \sinh z = -\left[(z-\pi i) + \frac{(z-\pi i)^3}{\lfloor 3} + \frac{(z-\pi i)^5}{\lfloor 5} + \cdots\right] \qquad |z-\pi i|<\infty$$

$$= -\sum_{n=0}^{\infty} \frac{(z-\pi i)^{2n+1}}{\lfloor 2n+1} \qquad |z-\pi i|<\infty$$

(V) $f(z) = \dfrac{e^{z}}{1+e^{z}}$ about $z_0=0$

$$= \frac{1+e^{z}-1}{1+e^{z}} = 1 - \frac{1}{(1+e^{z})}$$

$$= 1 - \left[g(z_0) + \frac{g'(z_0)}{\lfloor 1} z + \frac{g''(z_0)}{\lfloor 2} z^2 + \cdots\right]$$

where $g(z) = \dfrac{1}{1+e^{z}}$ $\qquad$ where $z_0=0$

Just find out $g^{(n)}(z_0)$, where $z_0=0$.